



- PANDAS
- SEABORN
- PLOTLY
- PRATICS
- DATA INSIGHTS

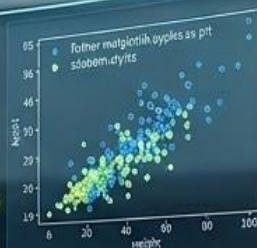
STARTING SOON

```
import matplotlib.pyplot as plt  
  
# Importing data  
data = pd.read_csv('data.csv')  
plt.scatter(data['x'], data['y'])  
plt.show()
```

```
# Data  
plt = matplotlib.pyplot  
plt = matplotlib.pyplot  
plt.show()
```

```
# Seaborn  
plt = sns.relplot(  
    data=data,  
    x='x',  
    y='y',  
    hue='category',  
    style='category',  
    facet_kws={'col': 'category'}  
)  
plt.show()
```

DATA VISUALIZATION



Data Visualization Camp using Python

22 – 25 June 2026



11:00 AM – 12:00 PM • 60 mins

Matplotlib Basics

Dr. Christodoss

- PANDAS
- PLOTLY
- PRATICS
- DATA INSIGHTS

Class begins shortly — please take a seat

STARTING SOON



Data Visualization
Camp using Python

22 – 25 June 2026



matplotlib

Python's #1 plotting library

Created in 2003 by John Hunter
to mimic MATLAB charts in Python

Used by:

- NASA scientists
- Medical researchers
- Financial analysts
- Data journalists
- You, in about 5 minutes!

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```
import matplotlib.pyplot as plt
# ...
plt.show()
```

```
plt = matplotlib.pyplot
plt = matplotlib.pyplot
plt.show()
```

```
import matplotlib.pyplot as plt
plt.show()
```

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matplotlib

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The Basic Recipe

```
# Step 1: Import the library
import matplotlib.pyplot as plt
```

```
# Step 2: Give it some data
x = [1, 2, 3, 4, 5]
y = [2, 4, 1, 5, 3]
```

```
# Step 3: Draw the chart
plt.plot(x, y)
```

```
# Step 4: Show it!
plt.show()
```

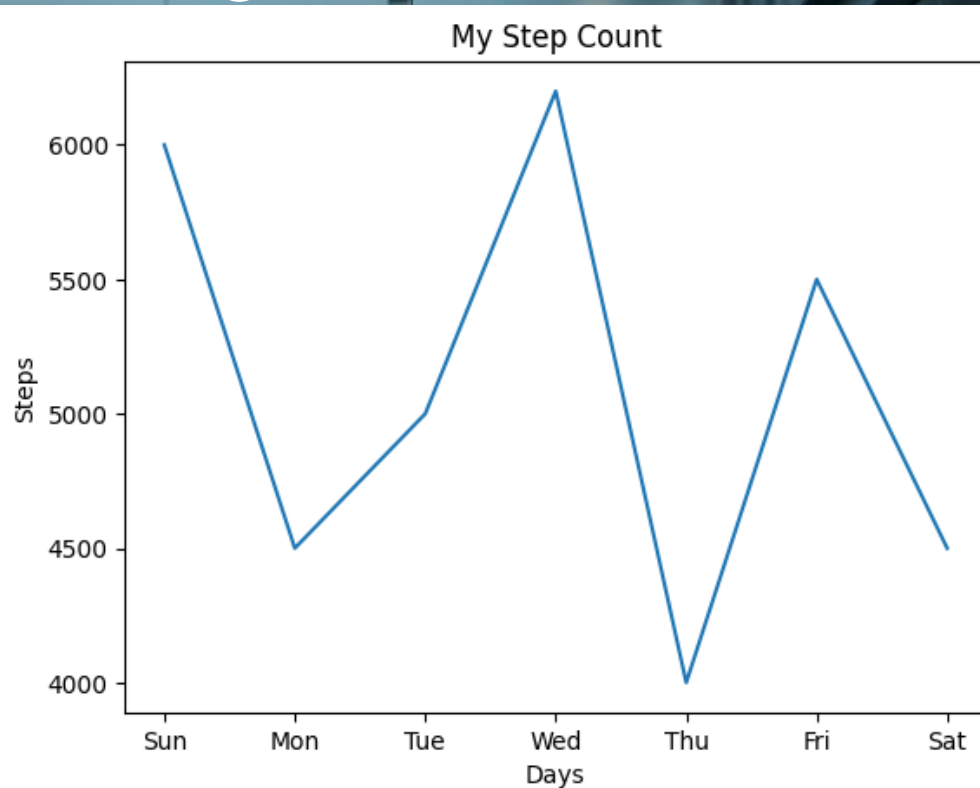
plt.plot() — Drawing Line Charts

Day 1

```
import matplotlib.pyplot as plt

# My step counts this week
days = ['Sun', 'Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat']
steps = [6000, 4500, 5000, 6200, 4000, 5500, 4500]

plt.plot(days, steps)
plt.title('My Step Count')
plt.xlabel('Days')
plt.ylabel('Steps')
plt.grid(True)
plt.show()
```



A chart without labels is confusing!

```
plt.title('My Chart')
```

Big title at the top of the chart

```
plt.xlabel('X-axis  
name')
```

Label for the horizontal (x) axis

```
plt.ylabel('Y-axis  
name')
```

Label for the vertical (y) axis

```
plt.grid(True)
```

Adds a helpful reference grid

```
plt.legend()
```

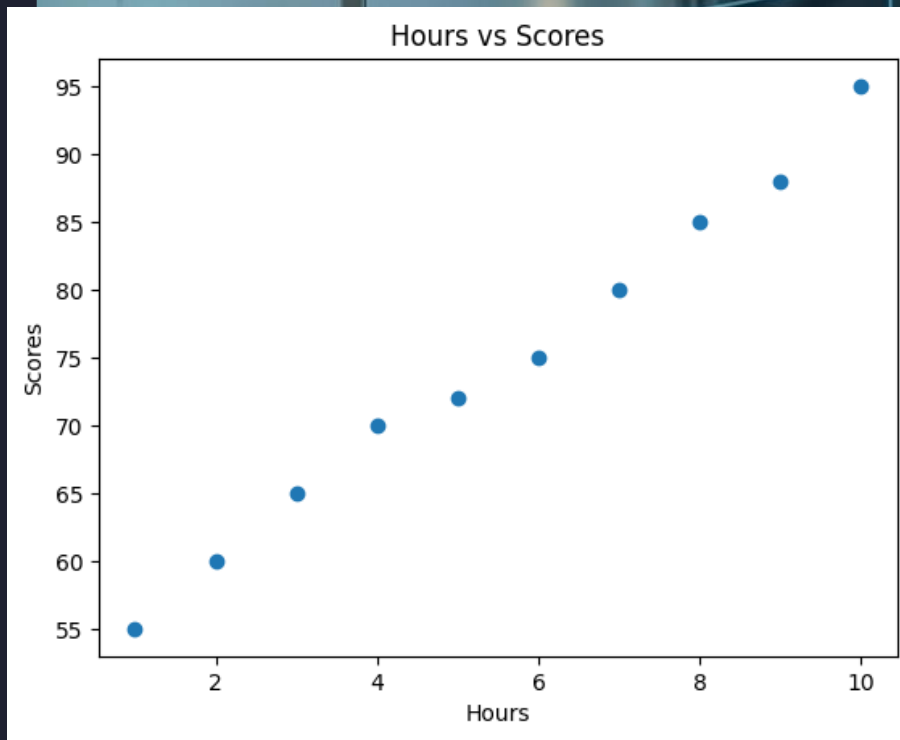
Shows what each line/color means

```
plt.plot(days, temps)    plt.title('Weekly Temps')    plt.xlabel('Day')    plt.ylabel('°F')  
plt.grid(True)    plt.legend()    plt.show()
```

```
import matplotlib.pyplot as plt

# Hours studied vs Test score
hours = [1,2,2,3,4,4,5,6,7,8,9,10]
scores= [55,60,65,70,72,75,
        80,85,88,95]

plt.scatter(hours, scores,
            color='blue',
            marker='o',
            s=100)
plt.xlabel('Hours')
plt.ylabel('Score')
plt.title('Hours vs Scores')
plt.show()
```



Putting It All Together

Day 1

```
import matplotlib.pyplot as plt
```

```
# Favorite snacks & calories
```

```
snacks = ['Pizza', 'Chips',  
          'Soda', 'Salad', 'Burger']
```

```
calories = [285, 150, 140,  
            30, 450]
```

```
plt.bar(snacks, calories,  
        color='steelblue')
```

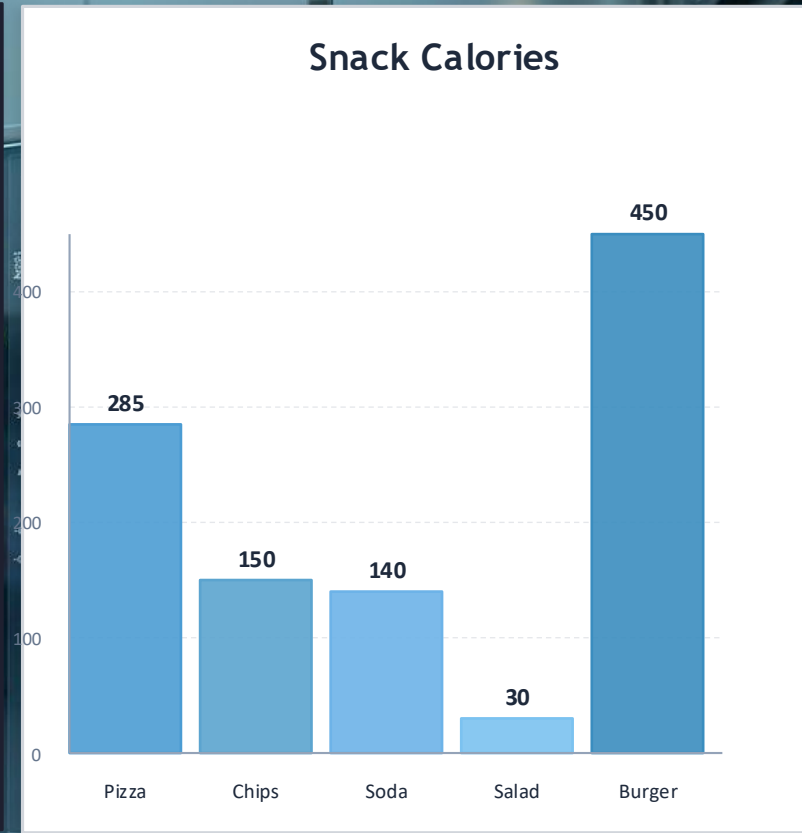
```
plt.title('Snack Calories')
```

```
plt.xlabel('Snack')
```

```
plt.ylabel('Calories')
```

```
plt.grid(axis='y')
```

```
plt.show()
```



Exercise 1: My Week in Steps

Day 1

```
import matplotlib.pyplot as plt

days = ['Mon', 'Tue', 'Wed',
        'Thu', 'Fri', 'Sat', 'Sun']

# ← Replace with YOUR step counts!
steps = [6000, 8000, 7500,
        9000, 5000, 11000, 4000]

plt.plot(days, steps,
         color='steelblue',
         marker='o', linewidth=2)

# Add your labels here:
plt.title('???')
plt.xlabel('???')
plt.ylabel('???')
plt.grid(True)
plt.show()
```

Your Tasks:

-  Type (or estimate) YOUR real step counts
-  Add a title with your name
-  Label both axes properly
-  Add `plt.grid(True)`
-  BONUS: Change `color='steelblue'` to your fav color
-  BONUS: Try `linestyle='--'` or `marker='*'`

15 minutes — Work in pairs! Share your chart when done

Exercise 2: Heights vs Shoe Sizes

Day 1

 Are taller people more likely to have bigger shoe sizes? Let's find out with data!

```
import matplotlib.pyplot as plt

# Collect data from 8 classmates
heights = [60, 62, 64, 65,
           67, 68, 70, 72]
shoe_sizes = [7, 7.5, 8, 8.5,
             9, 9.5, 10, 11]

plt.scatter(heights, shoe_sizes,
            color='purple', s=120,
            marker='D')

plt.title('Height vs Shoe Size')
plt.xlabel('Height (inches)')
plt.ylabel('Shoe Size')
plt.grid(True)
plt.show()
```

What to Try:

REQUIRED

Enter YOUR real height & shoe size

REQUIRED

Survey 4+ classmates — add their data

BONUS

Change marker to 'o', '*', or '^'

BONUS

Add a label for each person's name

REFLECT

What pattern do you see? Write it!



plt.scatter() is perfect for questions like 'Does X affect Y?'



Named Colors

	'steelblue'
	'coral'
	'gold'
	'mediumseagreen'
	'orchid'
	'tomato'
	'deepskyblue'
	'black'



Markers

'o'	Circle
's'	Square
'^'	Triangle Up
'D'	Diamond
'*'	Star
'x'	X mark
'+'	Plus
'.'	Dot



Line Styles

'-'	Solid
'--'	Dashed
'-.'	Dash-dot

```
plt.plot(x, y,
         color='coral',
         marker='*',
         linestyle='--',
         linewidth=3)
```

Your Matplotlib Cheatsheet

Day 1

```
import matplotlib.pyplot as plt
```

→ Load the library — always first!

```
plt.plot(x, y)
```

→ Draw a line chart connecting points

```
plt.scatter(x, y)
```

→ Draw individual dots (no lines)

```
plt.bar(labels, values)
```

→ Draw a vertical bar chart

```
plt.title('My Chart')
```

→ Add a title above the chart

```
plt.xlabel('X label')
```

→ Label for the x (horizontal) axis

```
plt.ylabel('Y label')
```

→ Label for the y (vertical) axis

```
plt.grid(True)
```

→ Add a reference grid to the chart



Lunch Break! Back at 1:30 PM for the Guided Design Lab

Questions

- PANDAS
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```
!report matplotlib.pyplot as plt
```

```
# Importing Libraries
```

```
gdc = causalrobotic("gdc")
```

```
plt =
```

```
plt = gdc.plotter("gdc", "gdc")
```

```
plt.plotter()
```

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plt.plotter()
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plt.plotter()
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plt.plotter()
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plt.plotter()
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plt.plotter()
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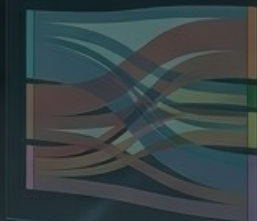
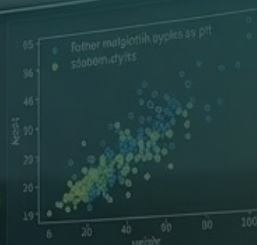
```
plt.plotter()
```

```
plt.plotter()
```

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plt.plotter()
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```
plt.plotter()
```

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